Generation and Reservoirs Statistics

December 29, 2024

PUBLIC UTILITIES COMMISSION OF SRI LANKA

1. Daily Generation Mix

December 29, 2024

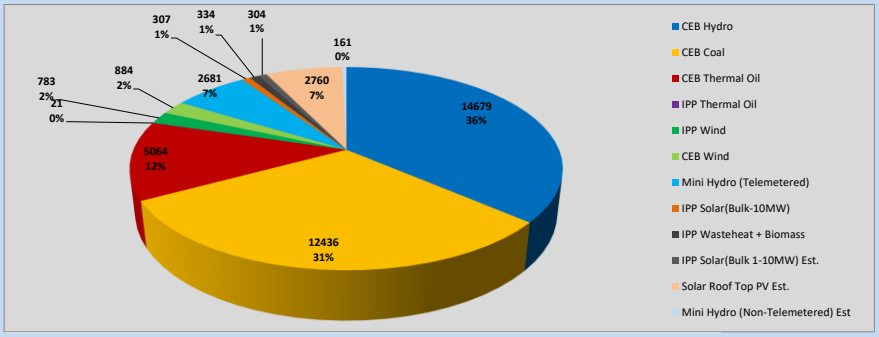


Table 01

	Generation (MWh)
CEB Hydro	14,679
CEB Coal	12,436
CEB Thermal Oil	5,064
IPP Thermal Oil	21
IPP Wind	783
CEB Wind	884
Mini Hydro (Telemetered)	2,681
IPP Solar (Bulk)	307
IPP Waste heat + Biomass	334
Total Generation (Excluding estimated figures)	37,189
* Estimated unserved energy	0
* Estimated Mini Hydro (Non telemetered)	161
* Estimated IPP Solar PV (Bulk 1-10MW)	304
* Estimated Solar Roof Top PV	2,760
Total Generation (Including estimated figures)	**40,414**

*** Estimated figures of CEB generation report**

1.1 Cumulative Dispatch - 2024

Table 02 - Current Month

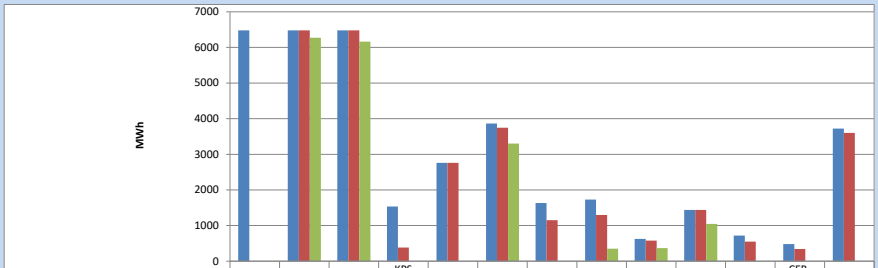
Category	Dispatch (GWh)	
CEB Hydro	662	50.62%
CEB Coal	308	23.57%
CEB Thermal Oil	44	3.40%
IPP Thermal	27	2.06%
IPP Wind	14	1.08%
CEB Wind	16	1.21%
Mini Hydro *	128	9.77%
IPP Solar *	92	7.02%
IPP Waste heat + BMP	10	0.80%
Total	**1,308**	

Table 03 - Current Year

Category	Dispatch (GWh)	
CEB Hydro	5,400	32.60%
CEB Coal	5,459	32.95%
CEB Thermal Oil	1,557	9.40%
IPP Thermal	762	4.60%
IPP Wind	398	2.40%
CEB Wind	382	2.30%
Mini Hydro *	1,382	8.34%
IPP Solar *	1,070	6.46%
IPP Waste heat	156	0.94%
Total	**16,566**	

*Including estimated contribution from non telemetered plants

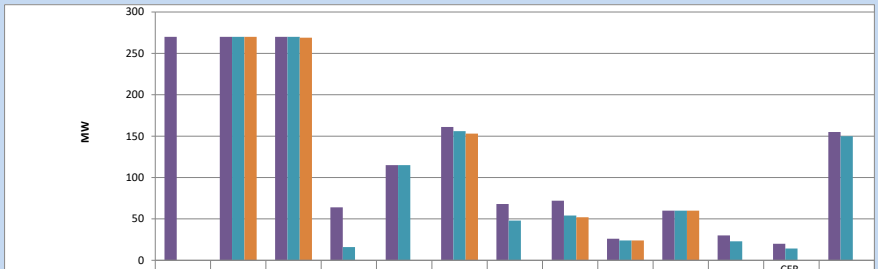
1.2 CEB owned Thermal Plant Dispatch December 29, 2024



	Coal I	Coal II	Coal III	KPS Small GTs	KPS GT 7	KCCP	Sapu A	Sapu B	Uthura Janani	Barge CEB	Dakunu Janani	CEB-Mathugama	KCCPS-2
Rated Daily Generation Capacity (MWh)	6480	6480	6480	1536	2760	3864	1632	1728	624	1440	720	480	3720
Available Generation (MWh)	0	6480	6480	384	2760	3744	1152	1296	576	1440	550	343	3600
Actual generation (MWh)	0.00	6273	6163	0	0	3302	0	349	368	1045	0	0	0

Available Generation is estimated based on plant availability at 6.00am on December 30, 2024

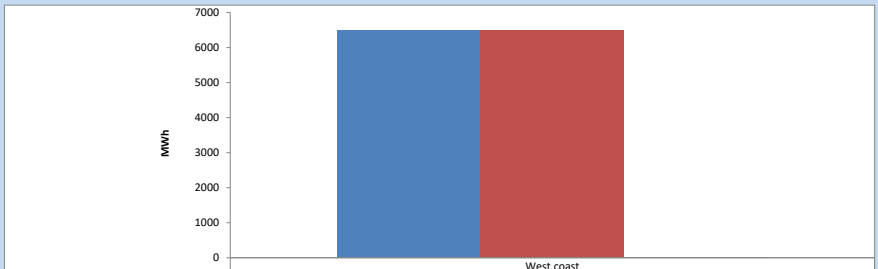
1.3 CEB owned Thermal Plant Loading at the Night Peak December 29, 2024



	Coal I	Coal II	Coal III	KPS Small GTs	KPS GT 7	KCCP	Sapu A	Sapu B	Uthura Janani	Barge CEB	Dakunu Janani	CEB-Mathugama	KCCPS -2
Installed Capacity (MW)	270	270	270	64	115	161	68	72	26	60	30	20	155
Available Capacity (MW)	0	270	270	16	115	156	48	54	24	60	23	14	150
Night Peak Load (MW)	0	270	269	0	0	153	0	52	24	60	0	0	0

Plant availability is recorded at 6.00 am on December 30, 2024

1.4 IPP owned Thermal Plant Dispatch December 29, 2024



	West coast
Rated Daily Generation Capacity (MWh)	6480
Available Generation (MWh)	6480
Actual generation (MWh)	21

Available Generation is estimated based on plant availability at 6.00am on December 30, 2024

1.5 IPP owned Thermal Plant Loading at the Night Peak

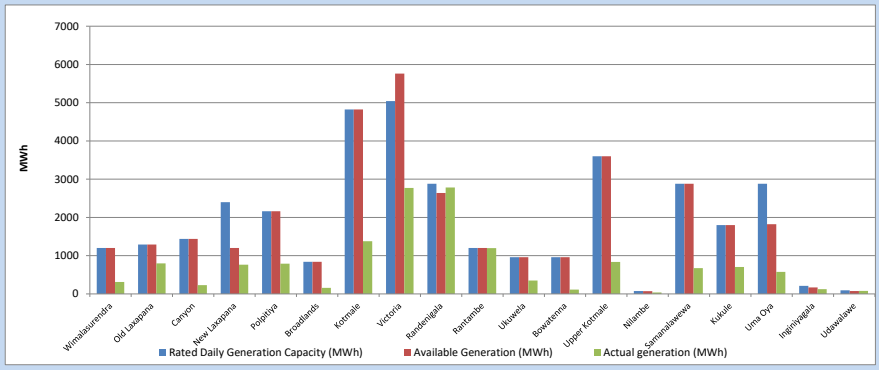


	West coast
Installed capacity (MW)	270
Available capacity (MW)	270
Night peak load (MW)	0

Plant availability is recorded at 6.00 am on December 30, 2024

1.6 Major Hydro Plant Dispatch

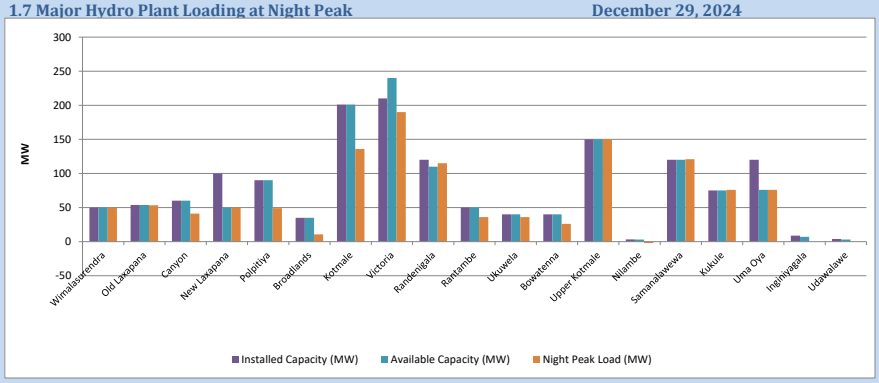
December 29, 2024



Available Generation is estimated based on plant availability at 6.00am on December 30, 2024

1.7 Major Hydro Plant Loading at Night Peak

December 29, 2024



Plant availability is recorded at 6.00 am on December 30, 2024

1.8 Summary of Major Plant performance

December 29, 2024

Table 04

Plant	Maximum Available Total Capacity	Plant Availability	Night peak Load	Plant Dispatch
	(MW)	(MW)	(MW)	(MWh)
Wimalasurendra	50	50	50	311
Old Laxapana	54	54	53	798
Canyon	60	60	41	228
New Laxapana	100	50	50	764
Polpitiya	90	90	49	790
Broadlands	35	35	11	158
Kotmale	201	201	136	1,380
Victoria	210	240	190	2,773
Randenigala	120	110	115	2,783
Rantambe	50	50	36	1,196
Ukuwela	40	40	36	351
Bowatenna	40	40	26	112
Upper Kotmale	150	150	150	836
Nilambe	3	3	-2	38
Samanalawewa	120	120	121	675
Kukule	75	75	76	706
Uma Oya	120	76	76	577
Inginiyagala	9	7	0	123
Udawalawe	4	3	0	79
Puttalam Coal I	270	0	0	0
Puttalam Coal II	270	270	270	6,273
Puttalam Coal III	270	270	269	6,163
KPS Small GTs	64	16	0	0
KPS GT 7	115	115	0	0
KCCP	161	156	153	3,302
Sapugaskanda A	68	48	0	0
Sapugaskanda B	72	54	52	349
Uthura Janani	26	24	24	368
Barge CEB	60	60	60	1,045
CEB-Hambantota	30	23	0	0
CEB-Mathugama	20	14	0	0
KCCPS -2	155	150	0	0
West Coast	270	270	0	21
Sobadhanavi	220	212	0	0
Total	3,602	2,924	2,263	37,188

Note-
Plant availability is the availability recorded at 6 am on December 30, 2024

1.9 Contribution to the Night Peak in MW

December 29, 2024

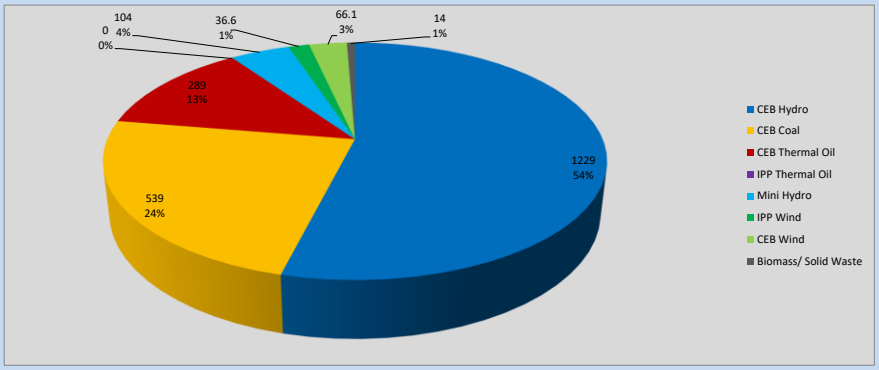


Table 05

CEB Hydro	1,229	MW
CEB Coal	539	MW
CEB Thermal Oil	289	MW
IPP Thermal Oil	0	MW
Mini Hydro (Telemetered)	104	MW
IPP Wind	36.6	MW
CEB Wind	66.1	MW
Biomass/ Solid Waste	14	MW

Recorded Peak Demand Data

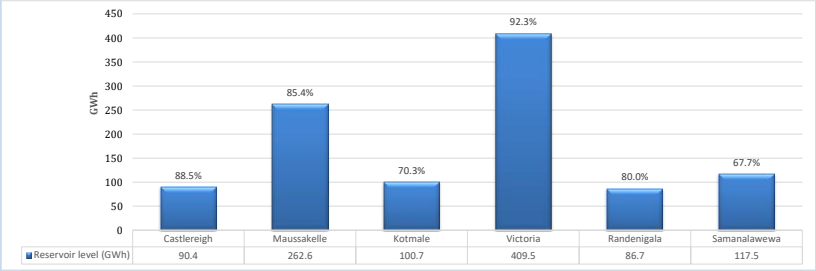
Table 06

Night Peak*	2,278	MW
Day Peak Maximum Demand	1,742	MW
Day Peak Minimum Demand	1,117	MW
Off Peak Minimum Demand	1,359	MW

Above figures are excluding contribution from roof top solar, non telemetered solar and mini hydro plants

1.10 Reservoir Levels -

as at 06.00 Hr on December 30, 2024



Total Reservoir Level 1,067 GWh
% of Total capacity 83.5%

1.11 Day Ahead Planned Demand Vs Actual Demand (Excluding non telemetered data)

December 29, 2024

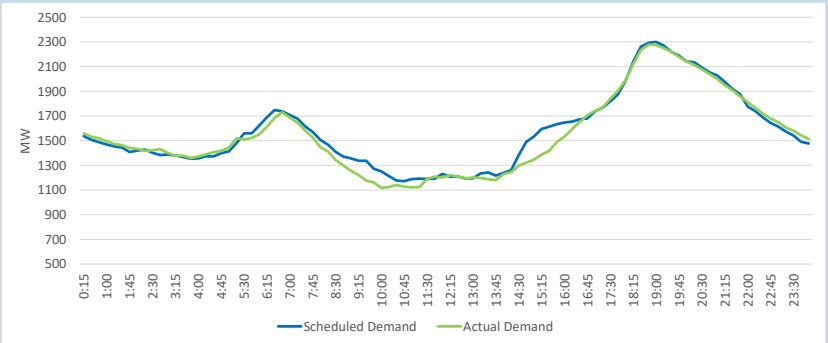
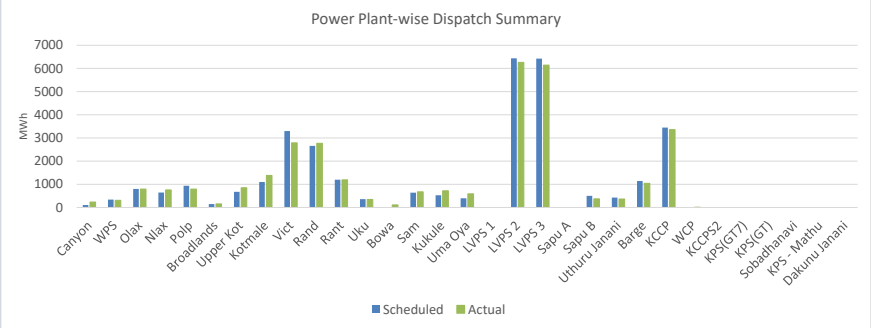


Table 07

Category	Scheduled Dispatch (MWh)	Actual Dispatch (MWh)	Deviation (MWh)
Major Hydro	13,824	14,679	855
CEB Coal	12,860	12,436	(424)
CEB Thermal Oil	5,520	5,064	(456)
IPP Thermal Oil	-	21	21
NCRE (excluding non telemetered power plants)	5,466	4,989	(477)
Total	**37,669**	**37,189**	**(480)**

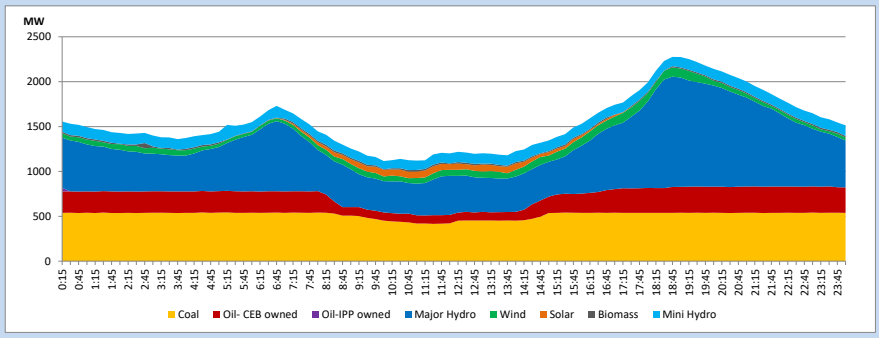
1.12 Power Plant-wise Dispatch Summary

December 29, 2024



1.13 Daily Load Curve

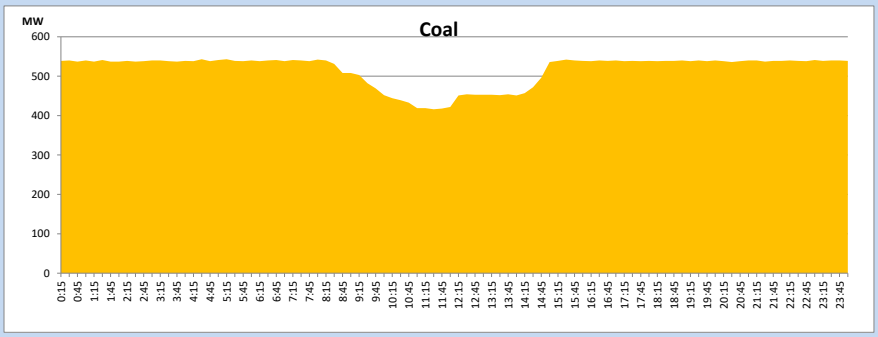
December 29, 2024



Solar and wind data is based on Telemetered Power Stations only

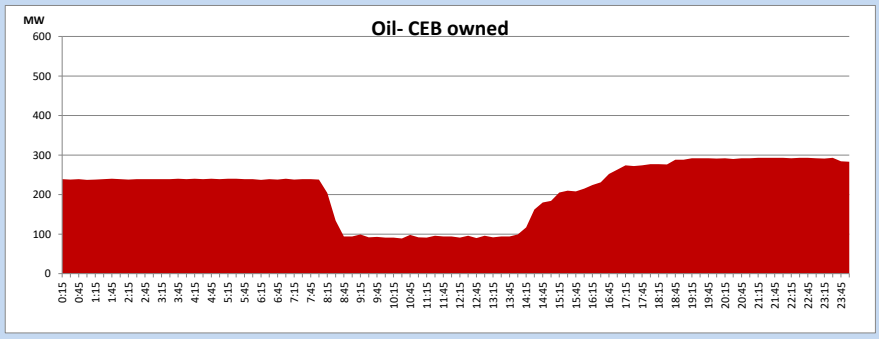
Coal Generation during

December 29, 2024

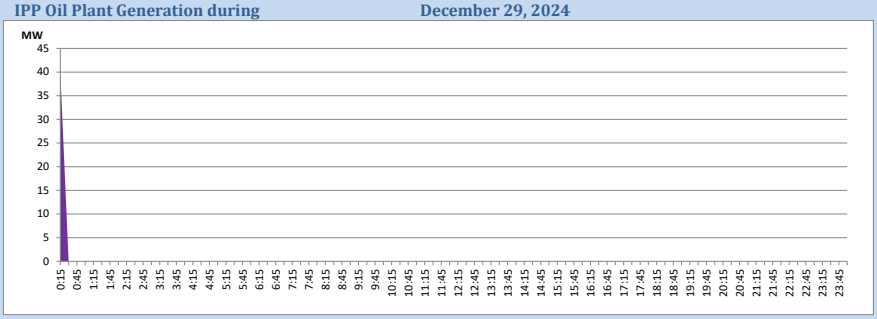


CEB Oil Plant Generation during

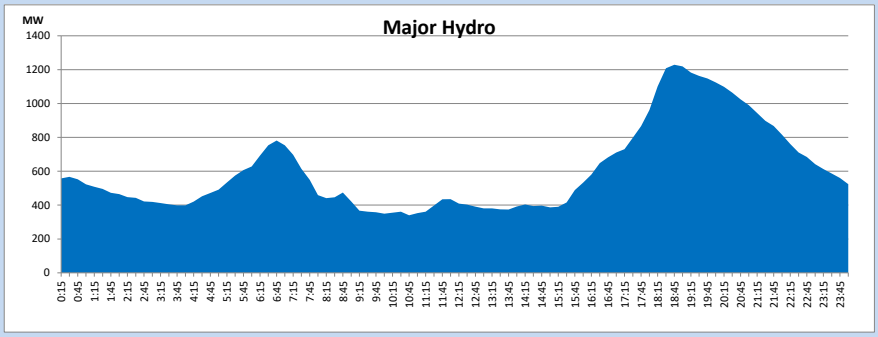
December 29, 2024



IPP Oil Plant Generation during December 29, 2024

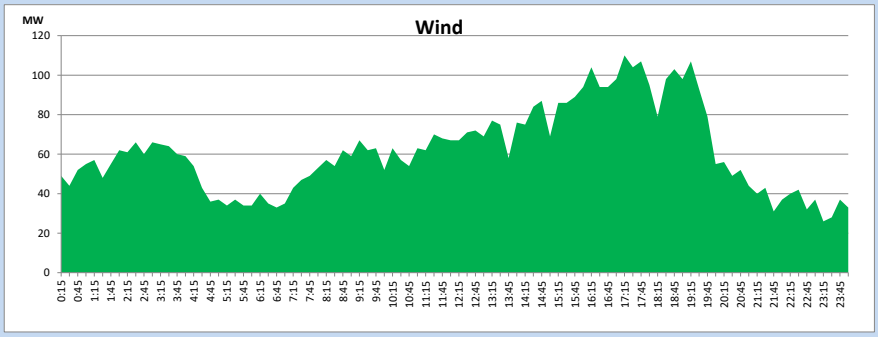


Major Hydro Generation during December 29, 2024



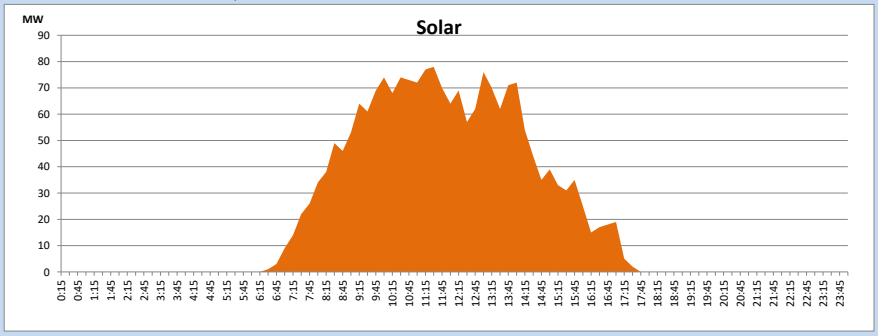
Wind Generation during December 29, 2024

Based on Telemetered Power Stations only



Solar Generation during December 29, 2024

Based on Telemetered Power Stations only



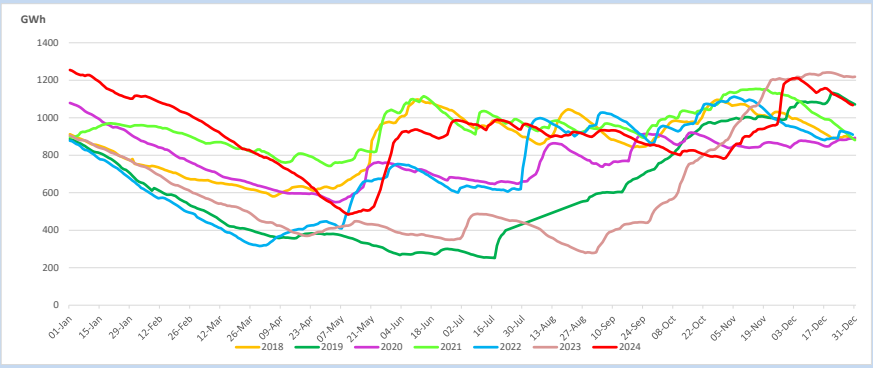
1.14 Major Incidents reported during the day December 29, 2024

1) Chunnakam - Kilinochchi 132kV cct 02 tripped from both ends at 15:19hrs due to the operation of distance protection. The cct was normalized at 15:31hrs.

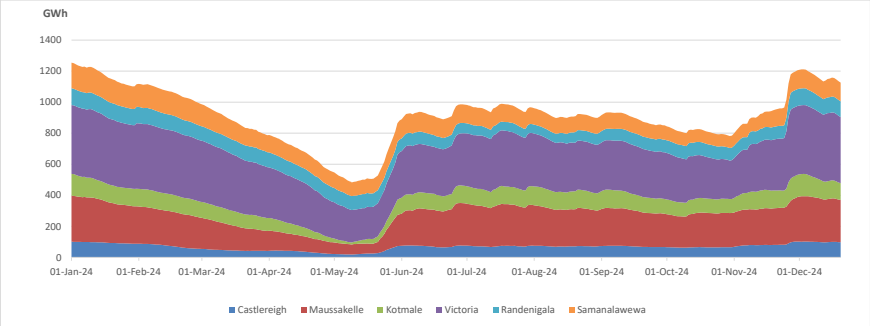
2) Uma Oya unit 01 generation is limited to 40MW from 15:52 hrs due to water leaking into the Governor oil sump.
3) Old Laxapana Stage - I tripped at 16:32hrs rejecting 30MW from the system, due to the penstock guard valve closing alarm.The Old Laxapana Stage - I resumed generation at 17:55hrs.

4) Chunnakam - Kilinochchi 132kV cct 01 tripped from both ends at 16:58hrs due to the operation of distance protection.
At the same time, Kilinochchi 132kV B/S CB tripped due to the operation of E/F protection. The B/S was normalized at 17:14hrs. The first attempt to energize Chunnakam - Kilinochchi 132kV cct 01 failed at 17:17 hrs, and the cct is yet to be normalized.

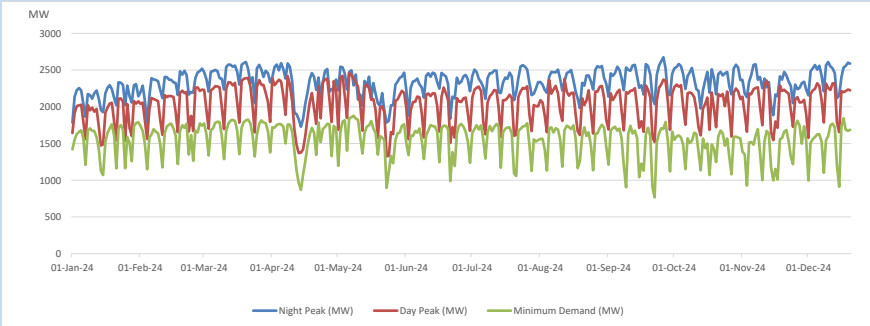
2. Comparison of Total Reservoir Storage Levels with Past Years



3. Variation of Major Hydro Reservoir Levels in the current year (GWh)

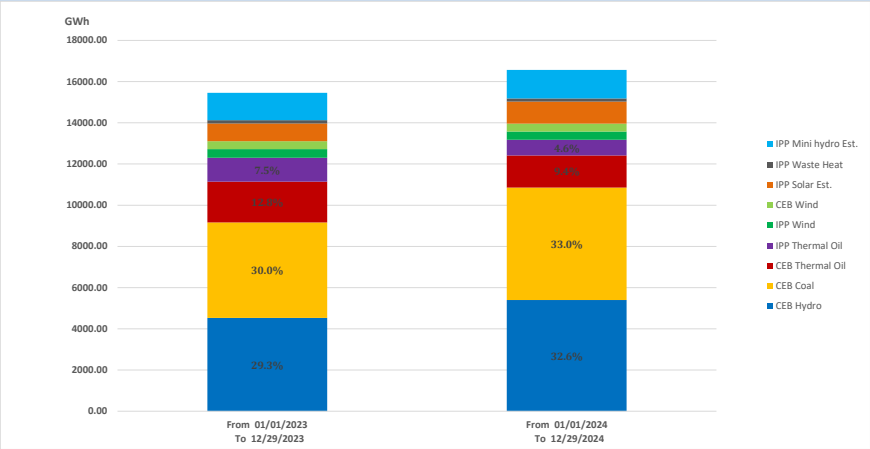


4. Variation of Demand during the current year



The above figures are excluding contribution from roof top solar, non telemetered solar and mini hydro plants

5. Cumulative Dispatch Comparison with Last Year



Cumulative dispatch

From 01/01/2023 To 12/29/2023	**15,456**	**GWh**
From 01/01/2024 To 12/29/2024	**16,566**	**GWh**

The above figures are including contribution from roof top solar, non telemetered solar and mini hydro plants)
Unserved energy due to power cuts has been excluded in 2023

Thermal Power Plant - Fuel types

Table 08

Power Station	Primary Fuel
CEB Thermal	
Sapugaskanda 1	Heavy Fuel
Sapugaskanda 2	Heavy Fuel
Kelanitissa Small Gas Turbines	Auto Diesel
GT 7 - Kelanitissa	Auto Diesel
Kelanitissa CCY	Naphtha or Diesel
Lakvijaya 1	Coal
Lakvijaya 2	Coal
Lakvijaya 3	Coal
Uthuru Janani	Heavy Fuel
Barge CEB	Furnace oil
KCCPS -2	Auto Diesel

Power Station	Primary Fuel
Private Thermal	
West Coast	Auto Diesel / Heavy Fuel
Sobadhanavi	Auto Diesel

6. Installed System Capcity

Table 09

	Installed Capacity (MW)
CEB Hydro	1,531
CEB Coal	810
CEB Thermal Oil	771
IPP Thermal Oil (West Coast & Sobadhanavi)	482
IPP Wind	163
CEB Wind	100
Mini Hydro	422
IPP Waste heat + Biomass	54
IPP Solar	137
Rooftop Solar (Ordinary) -CEB	495
Rooftop Solar (LT Bulk) -CEB	338
Rooftop Solar (HT Bulk)- CEB	104
Rooftop Solar - LECO	245

Data Source - Monthly Review Report Aug 2024 for NCRE installed capacities